Week 12: November 16-20
C 2.1 Chemical signaling (HL) and C 2.2 Neural signaling (SL and HL)

Subject/:الموضوع Biology **Gradeالصف : 12DP**

Lessonsالحصص	Learning targets أهداف التعلم	Assessment tools and strategies التقييم Formativeتكويني/ Summativeختامي / Differentiationالتمايز	Resources and links for Parents و المصادر الروابط المخصصة لأولياء الأمور
Lesson 1	I can explain and analyze how plants coordinate growth and responses to stimuli by interpreting seedling tropisms, tracing auxin distribution and action, comparing auxin–cytokinin interactions, and evaluating positive feedback in fruit ripening.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 7.2 Make learning relevant, authentic, and connected to students' backgrounds.	IB Biology, InThinking
Lesson 2	I can explain and analyze how cells communicate and respond to signalling chemicals by comparing receptor types, tracing signal transduction pathways, evaluating hormone actions, and explaining positive and negative feedback regulation.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 7.2 Make learning relevant, authentic, and connected to students' backgrounds.	IB Biology, InThinking
Lesson 3	I can explain and analyze how neurons transmit information by relating ion concentration gradients to resting potential, tracing action potentials along nerve fibres, comparing impulse speeds, and explaining neurotransmitter release at synapses.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 7.2 Make learning relevant, authentic, and connected to students' backgrounds.	IB Biology, InThinking
Lesson 4	I can analyze how neurons generate and integrate electrical signals by interpreting action-potential traces, explaining saltatory conduction and synaptic modulation, evaluating excitatory and inhibitory inputs, and relating neural activity to pain perception and consciousness.	2.5 Use multiple media to illustrate concepts (diagrams, videos, models). 3.3 Guide processing with explicit steps, chunking, and interactive models. 7.2 Make learning relevant, authentic, and connected to students' backgrounds.	IB Biology, InThinking

Homework **الواجبات**			
Teacher's notes/ Reminders **ملاحظات المعلم والتذكير**			

Week # 12 #الأسبوع

Technology Integration دمج التكنولوجيا	plant-growth models and digital seedling images to compare growth under uniform and lateral light annotated diagrams to trace auxin transport and visualize how a concentration gradient can produce differential cell elongation cell-signalling models to trace how an extracellular ligand activates a receptor and initiates an intracellular pathway digital animations or annotated diagrams help students visualize G-protein activation, second-messenger signalling, receptor tyrosine kinase activity, and the movement of steroid hormones into cells interactive neuron models to examine ion distributions across the membrane and investigate how changes in membrane permeability influence membrane potential animated action-potential sequences help students visualize electrical propagation along an axon
Cross-curricular links **الروابط بين المواد الدراسية**	Chemistry: Chemical signalling, concentration gradients, and the interactions between signalling molecules and target tissues. Mathematics: Measuring curvature, calculating mean shoot growth, graphing treatment results, and comparing growth rates. Environmental Science: Plant responses to light availability and implications for growing plants in shaded or densely planted environments. Chemistry: Molecular structure, polarity, solubility, ligand–receptor interactions, and chemical signalling. Mathematics: Interpretation of concentration–response graphs, comparison of experimental groups, and analysis of changes in pathway activity. Health Sciences: Hormone action, neurotransmission, reproductive physiology, and the effects of altered cell signalling. Microbiology: Bacterial communication, population density, and coordinated microbial behaviour. Chemistry: Ion gradients, electrochemical potential, diffusion, and membrane transport. Physics: Electrical potential differences, current, and propagation of electrical changes. Mathematics: Reading membrane-potential graphs, comparing numerical impulse speeds, and identifying changes over time.
External Assessment Integration دمج التقييمات الخارجية	Paper 1: Multiple-choice questions on positive phototropism, auxin distribution, auxin efflux carriers, cytokinin interactions, and ethylene-mediated positive feedback.

	Paper 2: Data-based questions requiring students to interpret seedling curvature or hormone-treatment results, explain concentration gradients and differential growth, compare developmental responses, and construct an explanation of positive feedback in fruit ripening. Paper 1: Multiple-choice questions on receptor specificity, hormone and neurotransmitter categories, membrane-potential changes, G proteins, epinephrine, tyrosine kinase receptors, intracellular receptors, and feedback. Paper 2: Structured questions requiring students to trace a signal-transduction pathway, compare receptor mechanisms, explain how steroid hormones affect gene expression, interpret signalling data, and evaluate evidence from experiments involving receptor inhibition or pathway disruption. Paper 1: Multiple-choice questions on resting potential, sodium–potassium pumping, action potentials, impulse speed, synapses, neurotransmitter release, and EPSPs. Paper 2: Structured questions requiring students to explain ion movement, trace the stages of synaptic transmission, interpret membrane-potential diagrams, and predict the effects of disrupted neurotransmitter release.
Wellbeing/ ATL Skills الصحة النفسية/ أساليب التعلم	Research & Thinking (ATL): Formulate predictions, interpret visual evidence, identify variables, and evaluate the quality of an experimental design. Communication: Use accurate terminology to explain hormonal signalling, concentration gradients, differential growth, and feedback loops. Collaboration: Compare seedling observations, justify predictions, and provide evidence-based feedback on experimental designs. Research & Thinking (ATL): Form hypotheses, predict outcomes when signalling components are blocked, interpret data, and distinguish evidence from assumptions. Communication: Explain receptor specificity and signal transduction using precise biological vocabulary, labelled diagrams, and cause-and-effect statements. Research & Thinking (ATL): Interpret models, predict the consequences of disrupting a pathway, and connect evidence to explanations. Communication: Explain the sequence of electrical and chemical signalling using accurate biological terminology.
Standards المعايير	C3.1.17 AHL Observations of tropic responses in seedlings C3.1.18 AHL Positive phototropism as a directional growth response to lateral light in plant shoots C3.1.19 AHL Phytohormones as signalling chemicals controlling growth, development and response to stimuli in plants

	C3.1.20 AHL Auxin efflux carriers as an example of maintaining concentration gradients of phytohormones C3.1.21 AHL Promotion of cell growth by auxin C3.1.22 AHL Interactions between auxin and cytokinin as a means of regulating root and shoot growth C3.1.23 AHL Positive feedback in fruit ripening and ethylene production C2.1.1 AHL Receptors as proteins with binding sites for specific signalling chemicals C.2.1.2 AHL Cell signalling by bacteria in quorum sensing C2.1.3 AHL Hormones, neurotransmitters, cytokines and calcium ions as examples of functional categories of signalling chemicals in animals C2.1.4 AHL Chemical diversity of hormones and neurotransmitters C2.1.5 AHL Localized and distant effects of signalling molecules C2.1.6 AHL Differences between transmembrane receptors in a plasma membrane and intracellular receptors in the cytoplasm or nucleus C2.1.7 AHL Initiation of signal transduction pathways by receptors C2.1.8 AHL Transmembrane receptors for neurotransmitters and changes to membrane potential C2.1.9 AHL Transmembrane receptors that activate G proteins C2.1.10 AHL Mechanism of action of epinephrine (adrenaline) receptors C2.1.11 AHL Transmembrane receptors with tyrosine kinase activity C2.1.12 AHL Intracellular receptors that affect gene expression C2.1.13 AHL Effects of the hormones oestradiol and progesterone on target cells C2.1.14 AHL Regulation of cell signalling pathways by positive and negative feedback C2.2.1 Neurons as cells within the nervous system that carry electrical impulses C2.2.2 Generation of the resting potential by pumping to establish and maintain concentration gradients of sodium and potassium ions C2.2.3 Nerve impulses as action potentials that are propagated along nerve fibres C2.2.4 Variation in the speed of nerve impulses C2.2.5 Synapses as junctions between neurons and between neurons and effector cells C2.2.6 Release of neurotransmitters from a presynaptic membrane C2.2.7 Generation of an excitatory postsynaptic potential --- C2.2.8 AHL Depolarization and repolarization during action potentials C2.2.9 AHL Propagation of an action potential along a nerve fibre/axon as a result of local currents C2.2.10 AHL Oscilloscope traces showing resting potentials and action potentials C2.2.11 AHL Saltatory conduction in myelinated fibres to achieve faster impulses

	C2.2.12 AHL Effects of exogenous chemicals on synaptic transmission (including cocaine) C2.2.13 AHL Inhibitory neurotransmitters and generation of inhibitory postsynaptic potentials C2.2.14 AHL Summation of the effects of excitatory and inhibitory neurotransmitters in a postsynaptic neuron C2.2.15 AHL Perception of pain by neurons with free nerve endings in the skin C2.2.16 AHL Consciousness as a property that emerges from the interaction of individual neurons in the brain
Inquiry/ TOK Question البحث والاستقصاء أسئلة TOK	*How can scientists infer the action of an invisible chemical signal from observable changes in an organism?* *How do scientists infer the mechanism of an invisible molecular process from observable changes in cells?* *To what extent can a model of ion movement explain the behaviour of a living neuron?*
Success Criteria معايير النجاح (Measurable)	لقد حقق الطالب/الطالبة أهداف التعلم من خلال: Lesson 1 MUST DOK 1 – Recall: Identify positive phototropism and describe the roles of auxin, cytokinin, and ethylene in the plant responses covered. SHOULD DOK 2/3 – Apply & Analyze: Explain how auxin distribution and cell elongation cause shoot bending, and how ethylene can contribute to positive feedback in fruit ripening. COULD DOK 4 – Evaluate & Create: Design a controlled investigation of seedling phototropism and create a justified model comparing auxin-mediated directional growth with ethylene-mediated positive feedback. Lesson 2 MUST DOK 1 – Recall: Identify the major signalling chemical categories and distinguish transmembrane receptors from intracellular receptors. SHOULD DOK 2/3 – Apply & Analyze: Trace an epinephrine signalling pathway and explain how receptor activation can lead to a cellular response through G proteins or other intracellular signalling mechanisms. COULD DOK 4 – Evaluate & Create: Design a controlled investigation of receptor-mediated signalling and create a justified model comparing membrane and intracellular receptor pathways, including the influence of positive and negative feedback. Lesson 3 MUST DOK 1 – Recall: Identify the roles of the sodium–potassium pump, action potentials, synapses, neurotransmitters, and EPSPs. SHOULD DOK 2/3 – Apply & Analyze: Trace an action potential to neurotransmitter release and explain how neurotransmitter binding produces an EPSP. COULD DOK 4 – Evaluate & Create: Construct and justify an integrated model of neural signalling, identifying how changes to ion gradients or neurotransmitter release would affect the pathway. Lesson 4 MUST DOK 1 – Recall: Identify the phases of an action potential, explain the role of myelin, and distinguish excitatory from inhibitory postsynaptic potentials.

	SHOULD DOK 2/3 – Apply & Analyze: Interpret an action-potential trace and explain how saltatory conduction, neurotransmitter reuptake, and synaptic summation influence neural signalling. COULD DOK 4 – Evaluate & Create: Use hypothetical experimental data to construct and justify a model linking action potentials, synaptic integration, and neural processing, while evaluating the limitations of using individual-neuron measurements to explain pain perception or consciousness.
National Identity Domain **مجالات الربط بالهوية الوطنية**	Conservation الحفاظ علي البيئة Global Understanding التفاهم العالمي Choose an item. المجال **Area: Sustainable agriculture, plant growth, food quality, and responsible management of natural resources in the UAE.** **Area: Scientific understanding of human health, cellular communication, and the biological mechanisms underlying health and disease in the UAE and globally.** **Area: Scientific understanding of the nervous system and its importance in health, sensory function, and communication between the brain and body.** التطبيق **How to Implement: Students examine how understanding plant growth responses can inform seedling placement and cultivation in environments with limited space or variable light. They also consider how knowledge of ethylene and fruit ripening can help inform post-harvest handling and reduce avoidable food waste. Students propose one evidence-based practice that supports more sustainable plant production or food management in the UAE.** **How to Implement: Present a case study showing how altered hormone signalling or receptor activity can affect normal cell function. Students explain the mechanism using scientific evidence and discuss why understanding molecular pathways contributes to informed health decisions and biomedical research. Emphasize that molecular mechanisms should be interpreted using evidence rather than assumptions about an individual's health.** **How to Implement: Use examples related to sensory and motor function to help students appreciate how neurological conditions can affect daily activities. Students discuss how accurate scientific knowledge can support empathy, accessibility, and respectful treatment of individuals experiencing neurological difficulties.**